

Rates of Change and Rectilinear Motion



Position, velocity, and acceleration

- 1 A particle moves along a line with position $s(t) = t^3 - 6t^2 + 9t$ (metres, seconds).
 - a Find the velocity function $v(t)$.
 - b Find the acceleration function $a(t)$.
 - c Find $v(2)$ and $a(2)$.
 - 2 For the particle above, find all times $t \geq 0$ when the particle is at rest ($v(t) = 0$).
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Speeding up and slowing down

- 3 Using the particle from above, determine whether the particle is speeding up or slowing down at $t = 2$ (compare the signs of $v(2)$ and $a(2)$).
 - 4 Determine whether the particle is speeding up or slowing down at $t = 0$.
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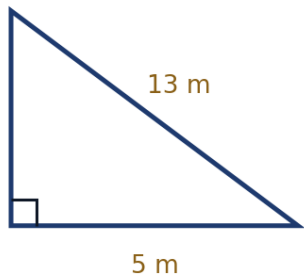
Total distance vs. displacement

- 5 For the particle above, find its displacement over $[0, 3]$.
 - 6 Using the fact that the particle changes direction at $t = 1$ (found earlier), find the total distance traveled over $[0, 3]$.
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Related rates

- 7 A spherical balloon is inflated so its radius grows at 2 cm/s. Find the rate of change of its volume when $r = 6$ cm, using $V = \frac{4}{3}\pi r^3$.

- 8 A 13 m ladder leans against a wall. The base slides away from the wall at 3 m/s. Find how fast the top of the ladder is sliding down when the base is 5 m from the wall.

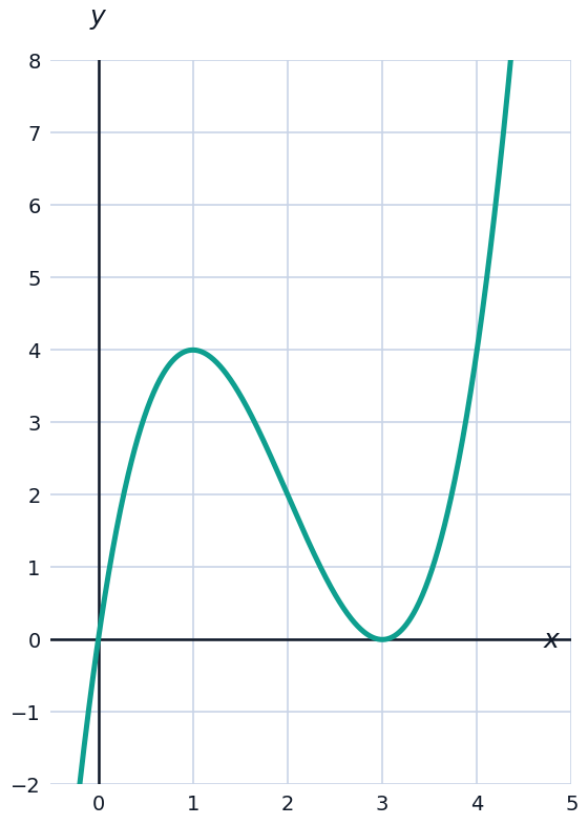


Motion from a velocity graph

- 9 A particle's velocity is $v(t) = t^2 - 4t$ (m/s) for $t \geq 0$. Find the times when the particle changes direction.
- 10 For the same particle, find the acceleration at $t = 1$ and state whether the particle is speeding up or slowing down at that instant.

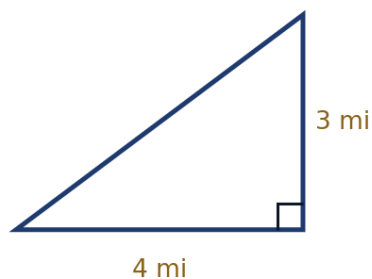
Visualizing position, velocity, and acceleration together

- 11 The graph shows the position function $s(t) = t^3 - 6t^2 + 9t$ from the first problem. Use the graph to identify the interval(s) where the particle is moving to the left (i.e. s is decreasing).



An applied related-rates capstone

- 12 A plane flies at a constant altitude of 3 miles, moving horizontally at 500 mph, directly away from a radar station on the ground. Find the rate at which the distance between the plane and the radar station is increasing when the plane is 4 miles horizontally from the station (directly above a point 4 miles from the station).



Motion with a velocity given as a formula

- 13 A particle's velocity is $v(t) = 6t - t^2$ (m/s) for $0 \leq t \leq 8$. Find the particle's position function $s(t)$ given $s(0) = 0$, then find its displacement over $[0, 6]$.
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A capstone rectilinear motion problem

- 14 A rocket's height is $h(t) = -5t^2 + 40t$ (meters, seconds) after launch. Find its maximum height, and the time at which it returns to the ground.